



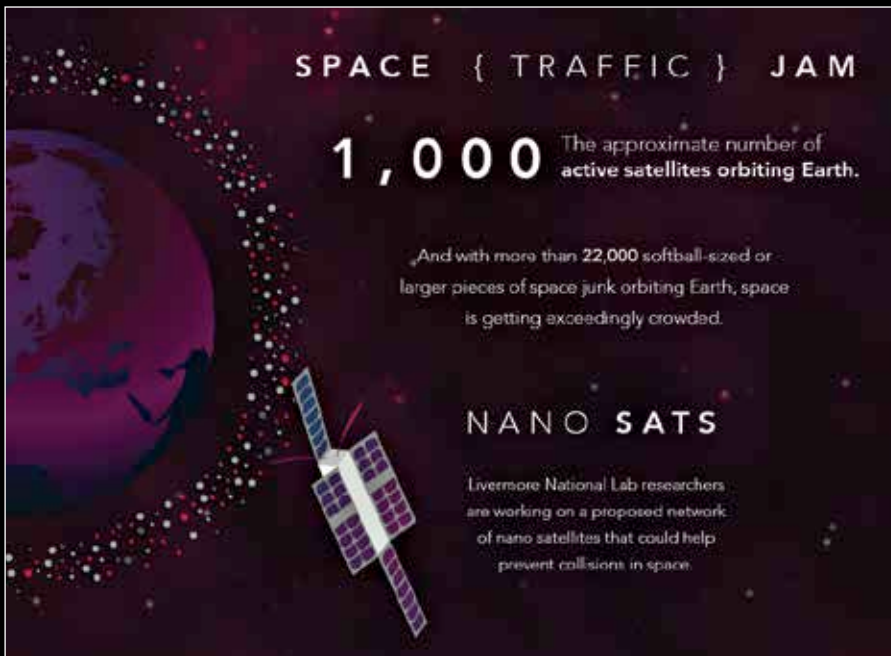
Fun Fact

If you drilled a tunnel straight through the Earth and jumped in, it would take you exactly 42 minutes and 12 seconds to get to the other side.



Lasering space junk

Japanese scientists are now proposing the use of lasers on the ISS to destroy space junk orbiting around the earth <http://digit.in/SpaceLasers>



Space junk orbiting the earth is nearly impossible to track for possible collisions

as perceived by humans appears to be infinite in its velocity but in reality it wasn't. Einstein's perception that the activation of the current, the view of the farmer and the jumping of the cows only "appeared" to be simultaneous to him, in his own context and very different from the farmer's context. Neither were wrong, just relatively different in their observations. This explanation by Einstein demonstrated that the concept of time is relative to the observer and thus was born the acclaimed Theory of Relativity. The cows in his dream were just condensed examples of cosmic phenomenon. A discovery so profound that it now exists as a cornerstone in all our cosmic ventures; from global positioning satellites and the search for Earth like planets to finding the true age of the Universe. Industries such as computing



A typical 3U CubeSat in groups of 18 costs only USD 30 million

and radio telescope wouldn't be possible without it. And to think - such an amazing discovery happened all because of a little inspiration from a dream about cows.

Traffic Jam Analysis

Another unlikely source for innovations in space research has come from observing mundane traffic jams. In the modern urban era, the problem of traffic has been a real pain. The issue of monitoring an ever increasing number of road vehicles in order to ensure safety and smooth operations has been a massive challenge. And even as the problem persists, many innovations have been utilised to combat the problem. Foremost of these solutions has been the use of traffic cameras. And it is this solution that has inspired space scientists to use the same concept in space.

As we all saw in the movie Gravity, space debris pose a massive risk to our well-being in space. Currently there are only 30 sensors in the Space Surveillance Network that track and monitor the 22,000 plus ball sized fragments orbiting Earth. This massive load of space junk is nearly impossible to track constantly and leads to over 10,000 false alarms of space collisions for every one genuine prediction. In order to improve the ability to track these objects more accurately, scientists

from the United States Energy Department's Lawrence Livermore National Labs (LLNL) have designed a system called the Space-Based Telescopes for Actionable Refinement of Ephemeris or STARE. This new system was designed in 2010 and is currently showing promise in testing phases. The system is built using a large network of nano-satellites which are low cost and easy to produce. Each of these nano-satellites is intended to serve as a traffic camera in space that monitors the orbits of thousands of debris particles and correlates them using advanced algorithms for actionable intelligence.

Each of these nano-satellites is in a shape slightly larger than a Rubik's cube and works in units of threes. These units are called the 3U CubeSats and are to be deployed into Earth orbit at seven kilometers per second. But thanks to the software on-board, the satellites can use their relative positions from the Earth and Sun, along with the visual tracking of fixed star constellations to know exactly where they are in orbit. From their unique vantage point they are then able to track orbital debris and give predictive analysis up to 48 hours in advance with a 100 fold increase in accuracy. All this from adapting something as simple and boring as traffic cameras.

Pop Culture Inspiration

The history of our species is littered with stories. Fantastical accounts of amazing



The magic of fiction inspired Robert Goddard to make his imagination a reality